EVERY BIVARIANT CLASS COMMUTES WITH THE CLASS OF A VIRTUAL NORMAL SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FBP (sources/stacks-chow/pinned-original-source.tex line 12206).

Allowed standard prerequisites (editorial).

- Testing identities of bivariant classes on fundamental classes. Same chapter, Lemma labelled lemma-bivariant-zero, cited at proof lines 12214-12216 and 12230-12232. Reduces to an integral X and allows the blowup reduction.
- The fundamental computation identifying the class with a top Chern class in the equal case. Same chapter, Lemma labelled lemma-gysin-fundamental, cited at proof lines 12223-12226. Handles the case Z = X.
- Decomposition of the class of a virtual normal sheaf along an effective Cartier divisor. Same chapter, Lemmas labelled lemma-gysin-decompose and lemma-gysin-agrees, cited at proof lines 12240-12245, and Lemma labelled lemma-gysin-bivariant for the Gysin homomorphism i^*. Writes the class as c_top(E) o i^* in the divisor case.
- Centrality of Chern classes among bivariant classes. Same chapter, Lemma labelled lemma-cap-commutative-chern, cited at proof lines 12226 and 12246-12248. Supplies the commutation with the top Chern class and with the product expression.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Same-chapter pointer into the author environment situation at source line 1372; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0FA2: sources/stacks-chow/pinned-original-source.tex, source lines 6200–6200. Same-chapter pointer into the author environment remark at source line 6200; invoked at proof line None of the counted proof.

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Source reference. Original lemma 02UC: sources/stacks-chow/pinned-original-source.tex, source lines 6585–6585. Same-chapter pointer into the author environment lemma at source line 6585; invoked at proof line 12215 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FBM: sources/stacks-chow/pinned-original-source.tex, source lines 12076–12076. Same-chapter pointer into the author environment lemma at source line 12076; invoked at proof line 12222 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02UA: sources/stacks-chow/pinned-original-source.tex, source lines 7581–7581. Same-chapter pointer into the author environment lemma at source line 7581; invoked at proof line 12223 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02OS: sources/stacks-chow/tags.txt, source lines 3527–3527. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 12231 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0B79: sources/stacks-chow/pinned-original-source.tex, source lines 6149–6149. Same-chapter pointer into the author environment lemma at source line 6149; invoked at proof line 12240 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FBL: sources/stacks-chow/pinned-original-source.tex, source lines 11994–11994. Same-chapter pointer into the author environment lemma at source line 11994; invoked at proof line 12243 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FBN: sources/stacks-chow/pinned-original-source.tex, source lines 12174–12174. Same-chapter pointer into the author environment lemma at source line 12174; invoked at proof line 12243 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be a scheme locally of finite type over S. Let $Z \subset X$ be a closed subscheme with virtual normal sheaf $\mathcal{N}$. Let $Y \to X$ be locally of finite type and $c \in A^*(Y \to X)$. Then c and $c(Z \to X, \mathcal{N})$ commute (Remark 0FA2).*

Proof. To check this we may use Lemma 02UC. Thus we may assume X is an integral scheme and we have to show $c \cap c(Z \to X, \mathcal{N}) \cap [X] = c(Z \to X, \mathcal{N}) \cap c \cap [X]$ in $\mathrm{CH}_*(Z \times_X Y)$.

If $Z = X$, then $c(Z \to X, \mathcal{N}) = c_{top}(\mathcal{N})$ by Lemma 0FBM which commutes with the bivariant class c, see Lemma 02UA.

Assume that Z is not equal to X. By Lemma 02UC it even suffices to prove the result after blowing up X (in a nonzero ideal). Let us blowup X in the ideal sheaf of Z. This reduces us to the case where Z is an effective Cartier divisor, see Divisors, Lemma 02OS,

If Z is an effective Cartier divisor, then we have

$$c(Z \to X, \mathcal{N}) = c_{top}(\mathcal{E}) \circ i^*$$

where $i^* \in A^1(Z \to X)$ is the Gysin homomorphism associated to $i : Z \to X$ (Lemma 0B79) and $\mathcal{E}$ is the dual of the kernel of $\mathcal{N}^\vee \to \mathcal{C}_{Z/X}$, see Lemmas 0FBL and 0FBN. Then we conclude because Chern classes are in the center of the bivariant ring (in the strong sense formulated in Lemma 02UA) and c commutes with the Gysin homomorphism i^* by definition of bivariant classes. □